

Tribhuvan University

BSc CSIT 7th Semester

Data Warehousing and Data Mining (CSC410)

Unit 2:

Introduction to Data Mining

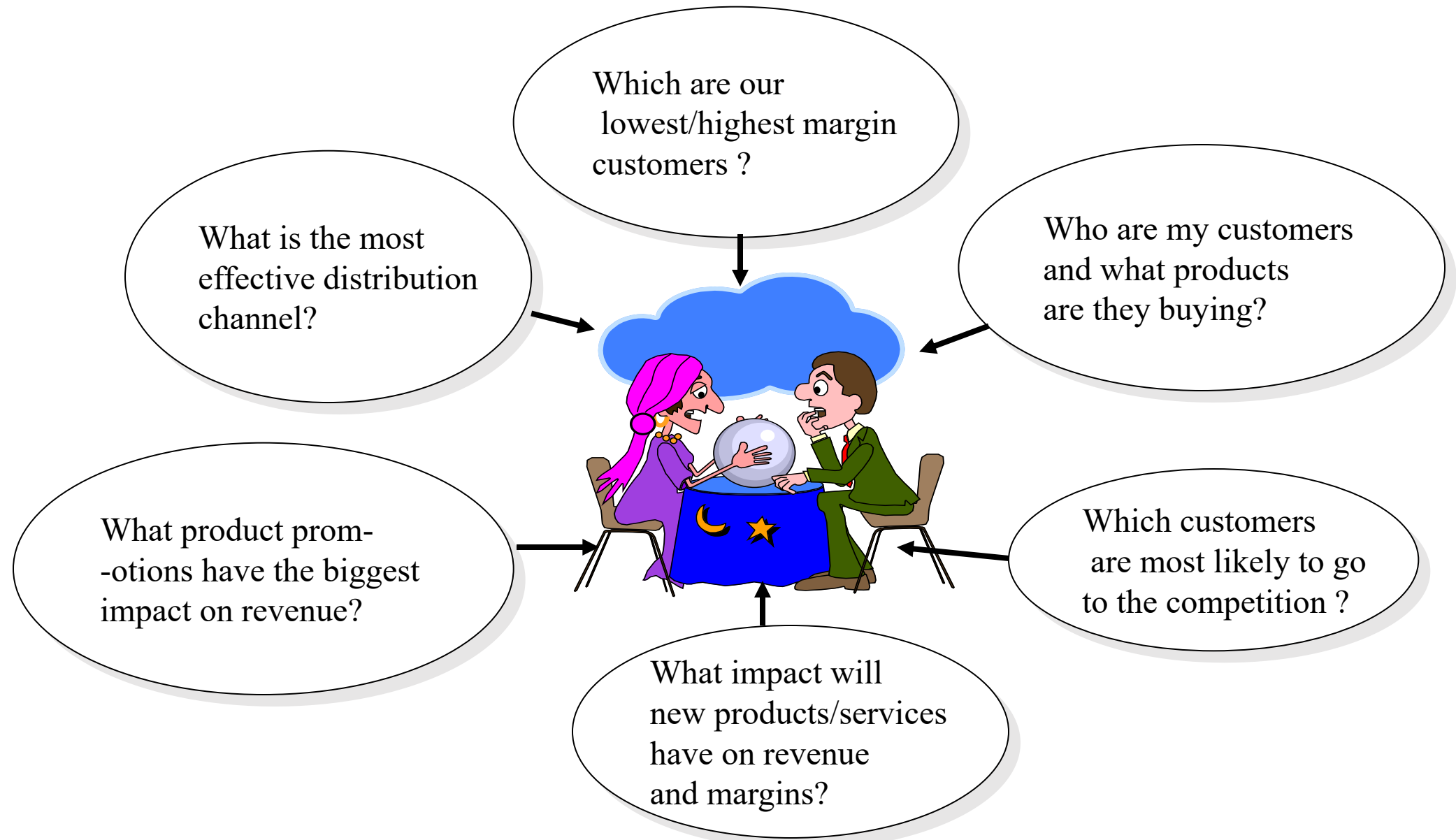
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Data Mining

- **Definition:** Data mining is the process of discovering patterns, trends, and insights from large datasets using statistical, mathematical, and computational techniques.
- **Purpose:** It extracts valuable knowledge from data to reveal hidden relationships and support decision-making, prediction, and optimization.
- **Application:** Businesses use data mining to understand customer behavior, improve marketing strategies, and enhance overall performance.

A producer wants to know....



Data, Data everywhere yet ...



- ❖ I can't find the data I need
 - data is scattered over the network
 - many versions, subtle differences
- ❖ I can't get the data I need
 - need an expert to get the data
- ❖ I can't understand the data I found
 - available data poorly documented
- ❖ I can't use the data I found
 - results are unexpected
 - data needs to be transformed from one form to other

We are **drowning in data**, but **starving for knowledge**

Foundations of Data Mining

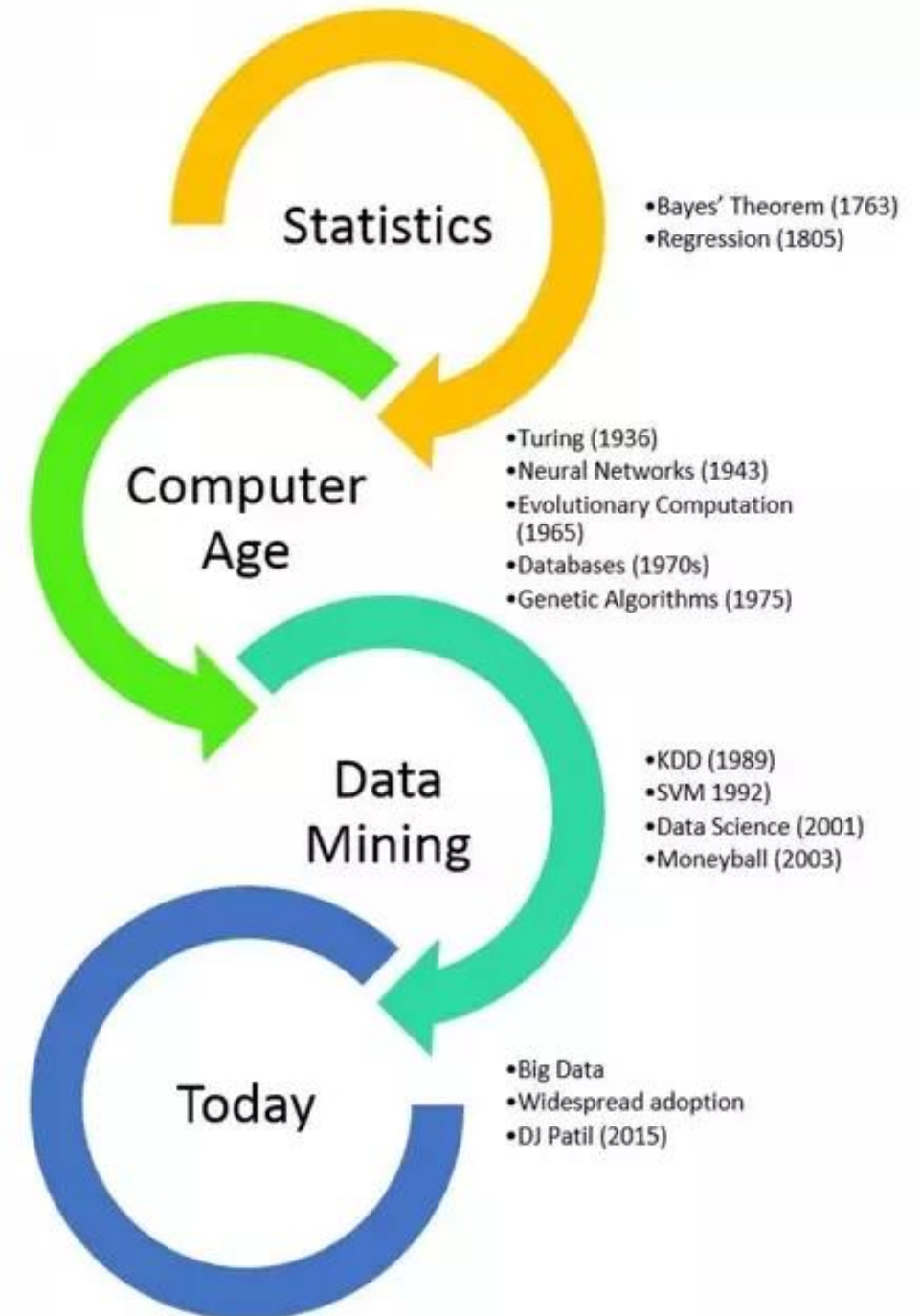
- **Data mining** can be viewed as a natural evolution of information technology.
- Data mining is an interdisciplinary field that draws techniques from several research areas.
- The major foundations of data mining are:
 - 1. Database Technology & Data Warehousing**
Provides mechanisms for data storage, integration, querying, and preprocessing.
 - 2. Statistics**
Supplies tools for data analysis, probability modeling, estimation, and hypothesis testing.
 - 3. Machine Learning & Artificial Intelligence**
Enables automatic learning, classification, clustering, prediction, and pattern discovery.
 - 4. Pattern Recognition & Information Retrieval**
Helps identify regularities in data and retrieve useful information from large datasets.
 - 5. Data Visualization & Domain Knowledge**
Supports interpretation of discovered patterns and ensures meaningful, actionable results.

Motivating Factors for Data Mining

- The primary motivation for data mining is the "**Data-Rich but Information-Poor**" paradox.
- While automated data collection tools have led to a massive data explosion, organizations often struggle to convert these vast archives into actionable knowledge
- Data mining has emerged to address the limitations of traditional data analysis techniques in handling modern, large-scale datasets.
- The following factors explain the motivation behind the development and adoption of data mining:
- **Data Explosion & Complexity:** Huge, high-dimensional, heterogeneous, and distributed datasets cannot be handled by traditional analysis methods.
- **Technological Advancements:** Affordable high-performance computing and mature data mining algorithms enable scalable and reliable analysis.
- **Business & Scientific Needs:** Data mining supports automated decision-making, fraud detection, and knowledge discovery.
- **Competitive Advantage:** Enables prediction, personalization, and strategic improvement through data-driven insights.

Origins of Data Mining

1. **Bayes' Theorem (1763):** This statistical formula describes how to update the probabilities of a hypothesis when given new evidence or information.
2. **Regression (1805):** This statistical method is used to estimate the relationships between a dependent variable and one or more independent variables to make predictions or understand causality.
3. **Turing (1936):** Alan Turing laid the theoretical foundation for modern computing by proposing the concept of the universal Turing machine, a mathematical model of a general-purpose computing machine.
4. **Neural Networks (1943):** Warren McCulloch and Walter Pitts proposed the first formal model of an artificial neuron, a simple computational device that uses binary inputs and outputs to perform logical functions.
5. **Evolutionary Computation (1965):** This field involves algorithms inspired by biological evolution, such as genetic algorithms, where the best "programs" are merged to evolve into better solutions.
6. **Databases (1970s):** This period saw the rise of systems for storing, managing, and organizing large amounts of data, providing the foundation for later data analysis efforts.
7. **Genetic Algorithms (1975):** These are search heuristics that mimic the process of natural selection, used to generate useful solutions to optimization and search problems.



Origin of Data Mining

- 34. **KDD (Knowledge Discovery in Databases) (1989):** This term was first used in a workshop to describe the multi-step process of discovering previously unknown and useful patterns from large databases, of which data mining is a key step.
- 35. **SVM (Support Vector Machines) (1992):** This supervised machine learning method is used for classification and regression tasks by constructing optimal hyperplanes that separate data points into different classes.
- 36. **Data Science (2001):** The term gained prominence as a multidisciplinary field integrating methods from statistics, machine learning, and databases to extract knowledge from data.
- 37. **Moneyball (2003):** This book and subsequent movie popularized the use of objective statistical analysis (sabermetrics) in professional baseball to make data-driven decisions and identify undervalued talent.
- 38. **Big Data:** This refers to the era characterized by extremely large datasets that can be analyzed computationally to reveal patterns, trends, and associations, especially relating to human behavior and interactions.
- 39. **Widespread adoption:** Data analysis techniques have moved from specialized fields into a wide range of industries and everyday applications, including healthcare, finance, and marketing.
- 40. **DJ Patil (2015):** Appointed as the first U.S. Chief Data Scientist, Patil led efforts to establish data policies across the federal government and launched initiatives like the Precision Medicine Initiative.

Functionalities of Data Mining

- **Class Description**

Summarizes data characteristics.

Example: Finding average income and spending behavior of customers.

- **Frequent Pattern & Association Mining**

Discovers relationships among items.

Example: Customers who buy bread often buy butter (market basket analysis).

- **Classification**

Assigns data into predefined classes.

Example: Classifying emails as spam or non-spam.

- **Prediction (Regression)**

Predicts continuous values.

Example: Predicting house prices or future sales

- **Clustering**

Groups similar data without labels.

Example: Customer segmentation based on purchasing behavior.

- **Outlier (Anomaly) Analysis**

Detects unusual or abnormal data.

Example: Identifying fraudulent credit card transactions.

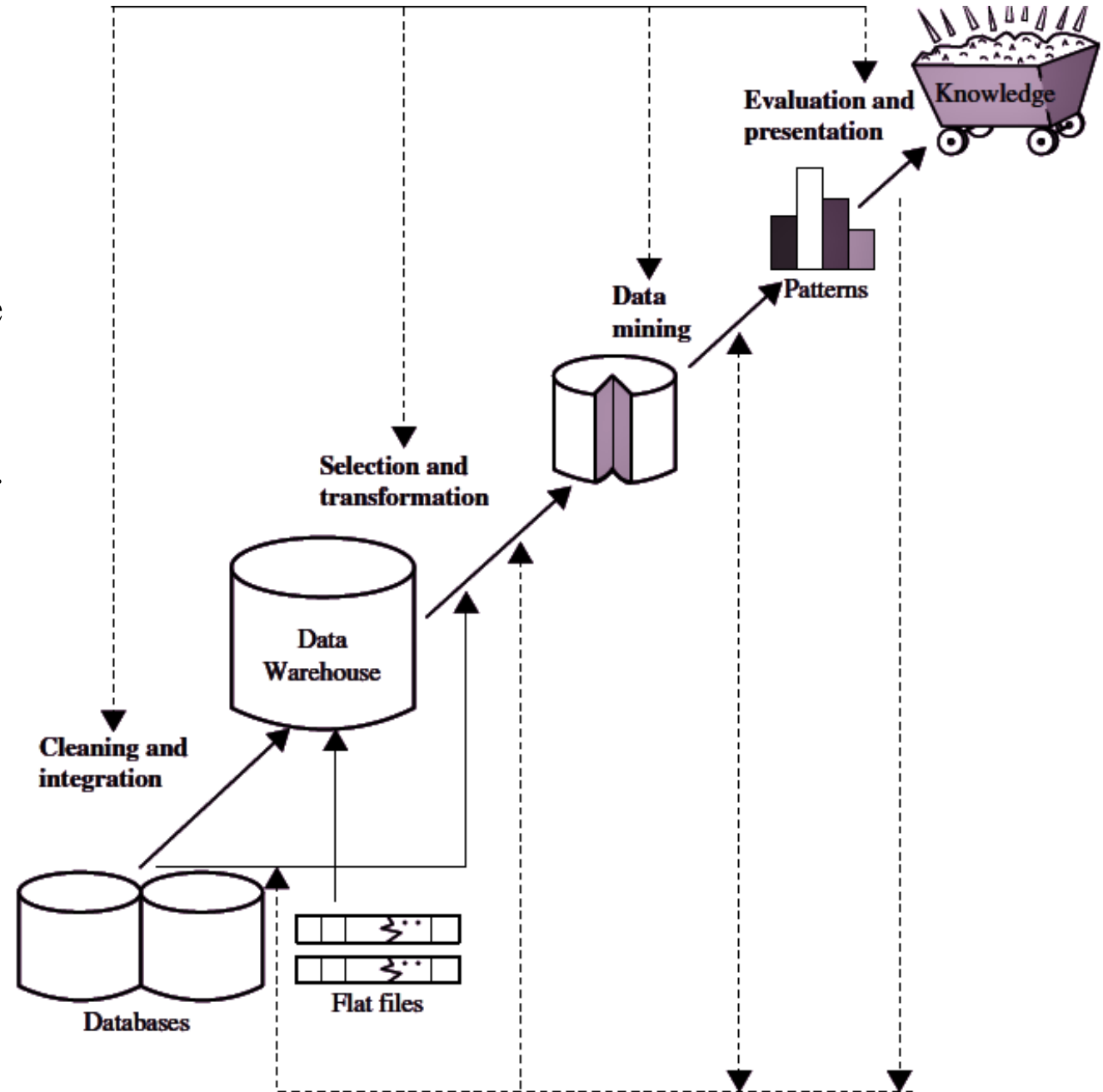
- **Evolution & Trend Analysis**

Analyzes data changes over time.

Example: Studying sales trends over multiple years.

KDD (Knowledge Discovery in Databases)

- KDD is the overall process of discovering useful, valid, novel, and understandable patterns from large datasets.
- Data mining is a **core step** within the KDD process.



Knowledge Discovery Steps

1. **Data Cleaning:** Removes noise, missing values, and inconsistencies from data.
2. **Data Integration:** Combines data from multiple sources into a unified dataset.
3. **Data Selection:** Selects relevant data required for analysis.
4. **Data Transformation:** Converts data into suitable formats (normalization, aggregation).
5. **Data Mining:** Applies algorithms to extract patterns and knowledge.
6. **Pattern Evaluation:** Identifies interesting and useful patterns.
7. **Knowledge Presentation:** Visualizes and presents discovered knowledge using charts, rules, or reports.

Data Mining Techniques

- In data mining, different kinds of knowledge or patterns can be discovered depending on the objective, data types, and mining techniques.
- Association Rules mining
- Classification
- Clustering
- Regression
- Outlier Detection (Anomaly Detection)

Data Mining Techniques

- **Association**

- Association is one of the best-known data mining technique.
- In association, a pattern is discovered based on a relationship between items in the same transaction.
- For example, when a female retail shopper buys a handbag, she is likely to buy shoes.

- **Classification**

- Data mining can partition the data so that the different classes or categories can be identified based on combination of parameters.
- For example, customers in the supermarket can be categorized into discount-seeking shoppers, shoppers in a rush, loyal regular shoppers, shoppers attached to name brands and infrequent shoppers.
- Classification is a classic data mining technique based on machine learning.
- Classification method makes use of mathematical techniques such as decision trees, linear programming, neural network, and statistics to classify each item in a set of data into one of a predefined set of classes or groups.

Data Mining Techniques

- **Clustering**

- Clustering is a data mining technique that makes a meaningful or useful cluster of objects which have similar characteristics using the automatic technique.
- The clustering technique defines the classes and puts objects in each class, while in the classification techniques, objects are assigned into predefined classes.
- For example: an entire population of treatment data on a disease may be divided into groups based on the similarity of side effects produced.

- **Regression**

- Regression uses existing values to forecast what other values will be.
- A regression task begins with a data set in which the target values are known.
- For example, a regression model that predicts children's height could be developed based on observed data for many children over a period of time. The data might track age, height, weight, developmental milestones, family history, and so on. Height would be the target, the other attributes would be the predictors, and the data for each child would constitute a case.

Association Rule Mining

- **Discover relationships between items in large datasets.**
 - In association, a pattern is discovered based on a relationship between items in the same transaction.
 - **Example 1:** When a female retail shopper buys a handbag, she is likely to buy shoes.
 - **Example 2:** "If a customer buys bread and butter, they are likely to buy milk."
- **Use Case:** Market Basket Analysis
- **Algorithm:** Apriori, FP-Growth

Classification

- **Assign data into predefined categories based on input features.**
 - **Example 1:** Classify an email → Spam or Not Spam.
 - **Example 2:** Classify whether a patient has a disease (e.g., Diabetes: Yes/No).
 - **Example 3:** Classify loan applicants as **High Risk** or **Low Risk**
 - **Example 4:** Classify students as **Pass** or **Fail**.
 - **Example 5:** Classify images into categories (e.g., **Dog, Cat, Horse**).
- **Algorithms:** Decision Trees, Naïve Bayes, SVM, Neural Networks

Clustering

- Group data into clusters where objects in the same cluster are more similar to each other than to those in other clusters.
- **Example 1: Customer segmentation in Marketing & Sales**
Group customers based on purchasing behavior, demographics, or browsing history.
Result: Segments like "Price-sensitive buyers", "Luxury shoppers", "Frequent buyers", etc.
- **Example 2: Document or News Article Clustering**
Group similar documents/articles together (e.g., all articles about politics, sports, or technology).
Result: Automatically generated topic-based clusters.
- **Algorithms:** K-Means, DBSCAN, Hierarchical Clustering

Regression

- **A predictive modeling technique** used to estimate the relationship between a **dependent variable** and one or more **independent variables**.
- **Purpose:**
 - Predict continuous outcomes
 - Understand influence of variables
 - Identify trends and relationships

Type	Description	Example
Linear Regression	Predicts value using a straight line	Salary vs Experience
Multiple Regression	Uses multiple predictors	House price vs Area, Rooms, Location
Polynomial Regression	Models curved relationships	Growth rate over time
Logistic Regression	(Technically classification) Used when target is binary	Loan approval (yes/no)

Outlier Detection

- Outlier detection is the process of **identifying data points that are very different** from the rest of the data.
- These unusual values are called **outliers**.
- Think of it as **spotting something strange or suspicious** in your dataset.
- They could be errors in data (e.g., typos, measurement issues)
- Or they could be important events, like fraud, machine failure, or rare diseases

Example: Outlier Detection

Credit Card Fraud Detection

- Imagine you work at a bank, and you're analyzing credit card transactions.
- Most people: Spend ₹500–₹5,000 daily and Shop locally
- Suddenly, you see: A transaction of ₹90,000 from a foreign country From a customer who usually spends ₹1,000 max

- **This is an outlier!**

It **doesn't follow the pattern** of normal behavior.

- You flag it for **fraud investigation**.

Data Mining Architecture

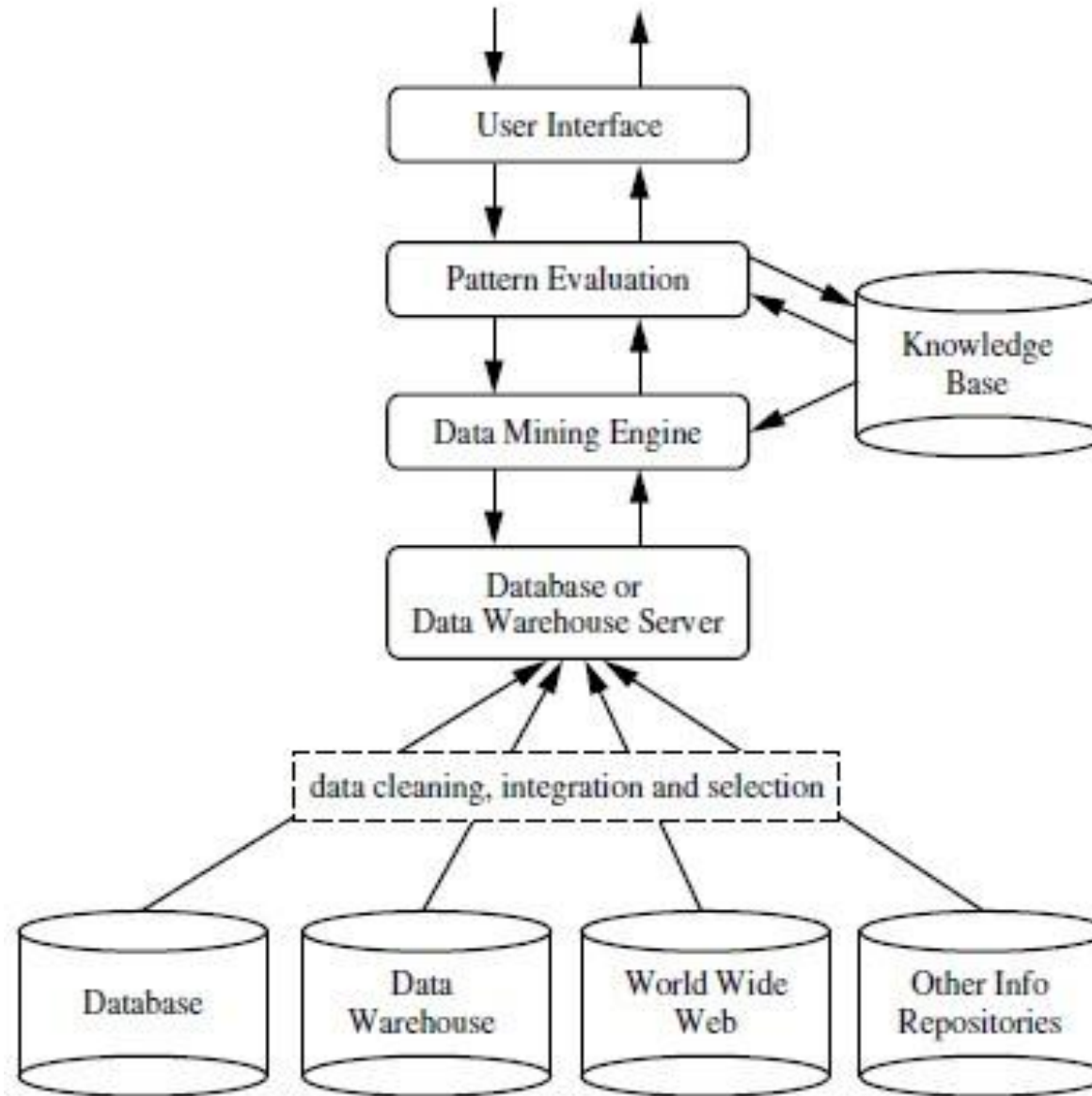


Fig. A Typical Data Mining Architecture

Data Mining Architecture

- **Data sources:** Data Sources consists of one or a set of databases, data warehouses, spreadsheets, Word Wide Web or other kinds of information repositories. Data cleaning and data integration techniques may be performed on the data.
- **Database or data warehouse server:** The database or data warehouse server is responsible for fetching the relevant data, based on the user's data mining request.
- **Knowledge base (KB):** A knowledge base is a centralized repository for information: a public library or a database of related information about a particular subject. A KB is not a static collection of information, but a dynamic resource that may itself have the capacity to learn, as part of an artificial intelligence (AI) expert system.

Data Mining Architecture

- **Data mining engine:** The data mining engine is the core component of any data mining system. It consists of a number of modules for performing data mining tasks including association, classification, characterization, clustering, prediction, time-series analysis etc.
- **Pattern evaluation module:** The pattern evaluation module is mainly responsible for the measure of interestingness of the pattern by using a threshold value. It interacts with the data mining engine to focus the search towards interesting patterns.
- **User interface:** This module communicates between users and the data mining system, allowing the user to interact with the system by specifying a data mining query or task, providing information to help focus the search, and performing exploratory data mining based on the intermediate data mining results. In addition, this component allows the user to browse database and data warehouse schemas or data structures, evaluate mined patterns, and visualize the patterns in different forms.

Data Objects and Attributes

- Data objects are individual entities or records that represent real-world items in a dataset.
Examples: student, customer, transaction, product.
- Attributes define the features of data objects.
- Each data object is described by multiple attributes.
 - For a student: Name, Age, Grade, Marks.
 - For a product: Product_ID, Price, Category, Stock.
 - For a customer: Customer_ID, Gender, Purchase Amount, City.
- Attributes can be numeric, categorical, binary, ordinal, discrete, or continuous.

Attribute Types

- Attributes are classified based on the kind of values they can take:
- **Nominal (Categorical) Attributes**
 - Represent names or labels with no order.
 - **Examples:** Gender (Male/Female), Blood Group (A, B, AB, O).
- **Binary Attributes**
 - Only two possible values.
 - **Examples:** Yes/No, True/False, 0/1.
- **Ordinal Attributes**
 - Values have a meaningful order but not exact differences.
 - **Examples:** Grades (A, B, C), Satisfaction Level (High, Medium, Low).
- **Numeric Attributes**
 - Represent measurable quantities.
 - Two types:
 - **Interval-scaled:** Equal intervals, no true zero (e.g., temperature in °C).
 - **Ratio-scaled:** Equal intervals with true zero (e.g., age, height, salary).
- **Discrete vs Continuous**
 - **Discrete:** Countable values (e.g., number of students).
 - **Continuous:** Infinite values within a range (e.g., height, weight, time).

Common Data Types

- **Numerical Data**

Data represented by numbers Consists of integers and floating-point numbers.

- This can further be divided into:
 - **Continuous:** Data that can take any value within a range. Examples include temperature, weight, height.
 - **Discrete:** Data that can only take specific values. Examples include counts of items, number of visits, etc.
- This type of data is commonly used in statistical analysis and is amenable to various data mining techniques such as regression, clustering, and predictive modeling.

Categorical Data

- Data represented by categories or labels that falls into categories or groups rather than numerical values
- These categories can be represented by labels, symbols, or names.
- This can further be divided into:
 - **Nominal**
 - **Ordinal**
- Categorical data is often used in descriptive data mining tasks, such as association rule mining and clustering, where the goal is to identify patterns or groupings within the data

Nominal Data

- Nominal data represent categories with no inherent order or ranking. Each category is distinct and unrelated to the others.
- Examples of nominal data include:
 - Gender (Male, Female)
 - Marital Status (Single, Married, Divorced)
 - Eye Color (Blue, Brown, Green)
 - Country of Residence (USA, Canada, UK)
- In nominal data, there is no meaningful numerical comparison between categories.
- For instance, "Male" is not greater or lesser than "Female"; they are simply different categories.

Ordinal Data

- Ordinal data represent categories that have a clear order or ranking.
- While the categories are distinct, they also have a relative position or hierarchy.
- Examples of ordinal data include:
 - Education Level (High School, Bachelor's, Master's, Ph.D.)
 - Income Level (Low, Medium, High)
 - Customer Satisfaction Rating (Poor, Fair, Good, Excellent)
- In ordinal data, there is a meaningful order between categories, but the magnitude of differences between categories may not be uniform.
- For example, the difference between "Poor" and "Fair" in customer satisfaction may not be the same as the difference between "Fair" and "Good".

Interval Scaled Data

- **Interval-scaled data** is a type of numerical data where the **differences between values are meaningful**, but the **zero point is arbitrary** (i.e., it does not indicate the absence of the quantity being measured).

Example	Explanation
Temperature in Celsius or Fahrenheit	0°C does not mean "no temperature"; 20°C is not "twice as hot" as 10°C.
Calendar Dates	The difference between 2000 and 2010 is meaningful (10 years), but there is no true "year zero".
IQ Scores	An IQ of 140 is higher than 120 by 20 points, but you can't say it's "twice as intelligent."

Interval Scaled Data

- We can:
 - Calculate the **difference** between two values
 - Use **mean, standard deviation, correlation**
 - **Add or subtract** values
- But we **cannot**:
 - Multiply/divide to interpret ratios (e.g., 40°C is not "twice as hot" as 20°C)

Ratio Scaled Data

- **Ratio-scaled data** is a type of quantitative data that has all the properties of interval-scaled data, **plus a true and meaningful zero point**.

Example	Explanation
Height or Weight	0 kg = no weight; 60 kg is twice as heavy as 30 kg
Age	0 years = not born; 20 years is twice as old as 10 years
Length or Distance	0 meters = no length; 100 m is twice as long as 50 m
Income	\$0 means no income; \$40,000 is double \$20,000
Kelvin Temperature	0 K = absolute zero (true absence of heat); 200 K is twice 100 K

Ratio Scaled Data

- With ratio-scaled data, we can:
 - Calculate **differences** (subtraction)
 - Compute **ratios** (division and multiplication)
 - Apply **all statistical techniques**: mean, median, mode, standard deviation, correlation, regression, etc.

Attribute Type		Description	Examples
Categorical (Qualitative)	Nominal	The values of a nominal attribute are just different names; i.e., nominal values provide only enough information to distinguish one object from another. (=, $\neq$)	zip codes, employee ID numbers, eye color, gender
	Ordinal	The values of an ordinal attribute provide enough information to order objects. ($<$, $>$)	hardness of minerals, { <i>good</i> , <i>better</i> , <i>best</i> }, grades, street numbers
Numeric (Quantitative)	Interval	For interval attributes, the differences between values are meaningful, i.e., a unit of measurement exists. (+, -)	calendar dates, temperature in Celsius or Fahrenheit
	Ratio	For ratio variables, both differences and ratios are meaningful. (*, /)	temperature in Kelvin, monetary quantities, counts, age, mass, length, electrical current

Statistical description of data

- In **data mining**, statistical description is used to **summarize and understand datasets** before applying mining algorithms.
- It helps identify patterns, trends, and anomalies in large volumes of data.
 - **Central Tendency** – Understand the typical value of an attribute.
 - *Mean*: Average value (e.g., average customer age).
 - *Median*: Middle value (robust to outliers).
 - *Mode*: Most frequent value (common category).
 - **Dispersion / Variation** – Understand the spread of data.
 - *Range, Variance, Standard Deviation*: Identify variability or consistency of data.
 - Helps in **normalizing data** for mining algorithms.
 - **Shape / Distribution** – Understand data patterns.
 - *Skewness*: Shows asymmetry in data (important for decision trees, regression).
 - *Kurtosis*: Shows whether data has extreme values (affects clustering or outlier detection).
 - **Graphical Summaries** – Visual exploration of data.
 - *Histograms, Boxplots, Scatter Plots*: Reveal trends, clusters, or anomalies.

Applications of Data Mining

- Data mining is widely used across industries to extract useful patterns and knowledge from large datasets.

Domain	Applications
Business/Marketing	Customer segmentation, sales forecasting, market basket analysis
Finance/Banking	Credit scoring, fraud detection, algorithmic trading
Healthcare	Disease prediction, patient profiling, drug discovery
Education	Student performance prediction, dropout analysis
Telecom	Network optimization, churn prediction, fraud detection

Applications of Data Mining

Domain	Applications
Government	Crime trend analysis, tax fraud detection, resource allocation
Web/Social Media	Recommender systems, sentiment analysis, web usage mining
Manufacturing	Predictive maintenance, quality control, supply chain optimization
Agriculture	Crop yield prediction, pest detection, soil analysis

Issues in Data Mining

- **Data Quality:** Presence of missing, noisy, or inconsistent data can affect mining accuracy.
- **Scalability:** Large and high-dimensional datasets require efficient algorithms to handle size and complexity.
- **Privacy & Security:** Sensitive data must be protected during collection, storage, and mining.
- **Heterogeneous Data:** Integrating structured, semi-structured, and unstructured data is challenging.
- **Distributed Data:** Data spread across multiple locations or organizations requires distributed mining techniques.

Thank You !